Patterns & Inductive Reasoning — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: What is inductive reasoning?*

Key Idea

Inductive reasoning: look at examples, find a **pattern**, and use it to **predict**.

1. Look → **2.** Describe the **rule** → **3.** Predict

☞ *Video: A number pattern*

Watch Me — Worked Example

$3,\ 7,\ 11,\ 15,\ \ldots$

Rule: **add 4** each time check **every** pair: $7-3,\ 11-7,\ 15-11$

Next term: $15+4=$ 19

10th term: $3+9\cdot 4=$ 39 start at 3, add 4 **nine** times

☞ *Video: A picture pattern*

Key Idea



$1,\ 4,\ 9,\ 16,\ \ldots$

These are **square numbers**: $1^2, 2^2, 3^2, 4^2$.

Next: $5^2=$ 25

☞ *Video: Conjecture and counterexample*

Key Idea

A **conjecture** is a statement you believe is true, based on a pattern.

Conjecture: "The sum of two numbers is always greater than each number."

A **counterexample**: $-2+5=$ 3, and $3<5$. So the conjecture is **false**.

One counterexample is enough to show a conjecture is false.

☞ *Video: Watch out*

Common Mistake

$2,\ 4,\ 8,\ \ldots$ Times 2? Next is 16. Or $+2, +4, +6$? Next is 14.

Two rules can fit the same first terms — that's why inductive reasoning gives a **conjecture**, not a proof.

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☞ *Video: Your turn*

Your Turn

Find the next two terms: $1,\ 3,\ 6,\ 10,\ \ldots$

The differences are $+2, +3, +4$, so next $__+5__$, then $__+6__$.

$10 + 5 = __15__$, $15 + 6 = __21__$

Notice

Remember: Look $\rightarrow$ describe the rule $\rightarrow$ predict. Check the rule on **every** pair. A pattern gives a **conjecture**; one **counterexample** shows it is false.

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